Combinatorics: Identities, Generating Functions, Catalan Numbers, and More

N-20240323

These exercises move from binomial identities and recurrences to convexity and graph walks. What first looks like a mixed problem set gradually becomes a catalogue of ways to encode counting: by subsets, generating functions, recursive decompositions, convex combinations, and matrix powers. The Catalan-number derivations and their combinatorial models follow Alexander Postnikov's MIT 18.212 algebraic-combinatorics lectures.[1]

§1.1 Binomial identities

Several binomial identities become clearer once their combinatorial meanings are made explicit. The basic identity

$$\sum_{i=0}^{n} \binom{n}{i} = 2^n$$

counts all subsets of an n-element set: choosing a subset means choosing its size i and then choosing i elements.

Pascal's identity

$$\binom{n}{i} + \binom{n}{i+1} = \binom{n+1}{i+1}$$

can be proved by fixing a distinguished element in an $(n+1)$-element set. An $(i+1)$-subset either contains the distinguished element or it does not. The two cases give the two terms on the left.

Vandermonde's identity

$$\sum_{i} \binom{m}{i}\binom{n}{r-i} = \binom{m+n}{r}$$

counts choosing r balls from two boxes with m and n balls. If i balls are chosen from the first box, then $r-i$ are chosen from the second.

Another useful identity is

$$\binom{n}{r}\binom{n-r}{k-r} = \binom{n}{k}\binom{k}{r}.$$

Both sides count a pair (R, K) with $R \subseteq K \subseteq [n]$, $|R| = r$, and $|K| = k$.

§1.2 Good subsets

A nonempty subset $A \subseteq \{1, 2, \ldots, n\}$ is called good if

$$|A| \leq \min_{x \in A} x.$$

[1] MIT 18.212, *Algebraic Combinatorics*, lecture notes.

Let a_n be the number of good subsets. The problem asks one to prove

$$a_{n+2} = a_{n+1} + a_n + 1.$$

A clean proof splits good subsets of $[n+2]$ into cases.

First, if $n+2 \notin A$, then A is just a good subset of $[n+1]$, giving a_{n+1} possibilities.

Second, if $A = \{n+2\}$, this gives one extra subset.

Third, suppose $n+2 \in A$ and $|A| \geq 2$. Remove $n+2$ and subtract 1 from every remaining element. If

$$A' = A \setminus \{n+2\},$$

then the condition becomes

$$|A'| + 1 \leq \min A'.$$

After shifting $B = \{x - 1 : x \in A'\}$, this is exactly

$$|B| \leq \min B,$$

so B is a good subset of $[n]$. This gives a_n possibilities. Hence

$$a_{n+2} = a_{n+1} + a_n + 1.$$

A summation proof fixes $|A| = k$. Every element must then be at least k, so the remaining data are chosen from a tail interval, and repeated Pascal identities yield the recurrence.

§1.3 Generating functions

For a sequence $(a_n)_{n\geq 0}$, its ordinary generating function is

$$f(x) = \sum_{n\geq 0} a_n x^n.$$

The note emphasizes a key principle: coefficients can encode counting problems.

For example, the number of nonnegative integer solutions of

$$x_1 + x_2 + \cdots + x_k = n$$

is the coefficient of x^n in

$$(1 + x + x^2 + \cdots)^k = \frac{1}{(1-x)^k}.$$

Using the binomial expansion,

$$\frac{1}{(1-x)^k} = \sum_{n\geq 0} \binom{n+k-1}{k-1} x^n,$$

so the number of solutions is

$$\binom{n+k-1}{k-1}.$$

The notes also use generating functions to separate parity or color constraints. A typical trick is to multiply several series, each recording the allowed choices for one box or color, and then read off the coefficient of the target power.

§1.4 Independent Trials and Generating Functions

For a nonnegative integer-valued random variable X, its probability generating function is

$$G_X(t) = \mathbb{E}[t^X] = \sum_{k\ge 0} \Pr(X = k)t^k.$$

If X, Y are independent, then

$$G_{X+Y}(t) = \mathbb{E}[t^{X+Y}] = \mathbb{E}[t^X]\mathbb{E}[t^Y] = G_X(t)G_Y(t).$$

For example, one Bernoulli(p) trial has generating function $1-p+pt$. Two independent batches of 2 and 3 such trials have respective generating functions $(1-p+pt)^2$ and $(1-p+pt)^3$. Their sum therefore has generating function

$$(1-p+pt)^5, \qquad \Pr(X+Y=2) = \binom{5}{2}p^2(1-p)^3.$$

This is the probabilistic analogue of ordinary generating-function convolution: independence turns counting or probability of a sum into multiplication of series.

§1.5 Catalan numbers

The note then turns to Catalan numbers. One model is the number of ways to triangulate a convex $(n+2)$-gon:

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$

For example, a hexagon has 14 triangulations.

Let h_n denote the number of triangulations of an $(n+2)$-gon. Choose the triangle incident with a fixed side. If its third vertex cuts the polygon into two smaller polygons, the numbers of triangulations multiply. Therefore

$$h_n = \sum_{i=0}^{n-1} h_i h_{n-1-i}, \qquad h_0 = 1.$$

If

$$H(x) = \sum_{n\ge 0} h_n x^n,$$

then the recurrence gives

$$H(x) = 1 + xH(x)^2.$$

Solving the quadratic and choosing the branch with $H(0) = 1$ gives

$$H(x) = \frac{1-\sqrt{1-4x}}{2x}, \qquad h_n = C_n.$$

This derivation explains why Catalan numbers appear in triangulations, parentheses, Dyck paths, and many noncrossing structures: all of them are recursively split into a left and a right part.

§1.6 Convex sets and convex hulls

The geometry part defines a convex set S by the condition that for any $x, y \in S$ and any $t \in [0,1]$,

$$tx + (1-t)y \in S.$$

Equivalently, S contains the line segment between any two of its points. The note contrasts convex and non-convex examples.

For finitely many points $x_1, \dots, x_k$, their convex hull is

$$\left\{ \sum_{i=1}^{k} t_i x_i : \ t_i \geq 0, \ \sum_{i=1}^{k} t_i = 1 \right\}.$$

Induction on k proves the claim: the case $k = 2$ is the definition of a segment, and the induction step writes the convex combination as a combination of one point with a convex combination of the remaining points.

§1.7 Separating a point from a closed convex set

Let $C \subset \mathbb{R}^n$ be nonempty, closed and convex, and let $x \notin C$. Because squared distance to x is continuous and grows without bound at infinity, it attains a minimum at some $y \in C$. For any $z \in C$, convexity gives $y + t(z-y) \in C$ for $0 \leq t \leq 1$. The function

$$\varphi(t) = \|x - y - t(z-y)\|^2$$

has its minimum at $t = 0$. Its right derivative therefore satisfies

$$0 \leq \varphi'(0) = -2\langle x - y, z - y \rangle.$$

Let $\ell(v) = \langle x - y, v \rangle$. Then every $z \in C$ obeys

$$\ell(z) \leq \ell(y) < \ell(x),$$

since $\ell(x) - \ell(y) = \|x - y\|^2 > 0$. Thus an affine hyperplane strictly separates the point x from C. The proof also explains the geometric role of projection onto a convex set.

For two arbitrary disjoint convex sets, a closest pair need not exist; additional closedness, compactness or other hypotheses must be checked before applying this proof to both sets.

§1.8 Planar graphs and Euler-type counting

Graph theory and planar configurations provide another counting mechanism: a planar drawing constrains vertices, edges, and faces through Euler-type formulas.

This connects naturally with the previous convex-geometry material: triangulations, planar graphs, polygon decompositions, and convex hulls are different languages for controlling incidence data.

§1.9 Graphs and adjacency matrices

A graph G consists of a vertex set $V = \{v_1, \dots, v_n\}$ and an edge relation. For a simple graph, the adjacency matrix $A(G) = (a_{ij})$ is defined by

$$a_{ij} = \begin{cases} 1, & \text{if there is an edge from } v_i \text{ to } v_j, \\ 0, & \text{otherwise.} \end{cases}$$

For a multigraph, a_{ij} records the number of edges from v_i to v_j.

The source quotes the standard theorem: the (i, j)-entry of $A(G)^\ell$ equals the number of walks of length ℓ from v_i to v_j. The proof is matrix multiplication. For $\ell = 2$,

$$(A^2)_{ij} = \sum_k a_{ik} a_{kj},$$

which counts all intermediate vertices v_k. The general case follows by induction.

§1.10 Catalan objects share a splitting rule

The parenthesizations of $n+1$ ordered factors and the triangulations of a convex $(n+2)$-gon are both counted by C_n. Choosing the final binary operation or the triangle incident with a fixed edge splits the remaining object into independent left and right pieces. Both satisfy

$$C_0 = 1, \qquad C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}.$$

A separate derivation uses Dyck paths. There are $\binom{2n}{n}$ paths with n upward and n downward steps. The reflection principle bijects those that ever cross below the horizontal axis with paths containing $n+1$ downward and $n-1$ upward steps, giving

$$C_n = \binom{2n}{n} - \binom{2n}{n+1} = \frac{1}{n+1}\binom{2n}{n}.$$

Recursion, generating functions and reflection now calculate the same number by distinct mechanisms; there is no need to restart the Catalan definition in several chapters of the same note.

§1.11 Adjacency powers count walks

For a finite undirected graph with adjacency matrix A, the entry

$$(A^k)_{ij}$$

counts length-k walks from vertex i to vertex j. For $k = 2$, this is

$$(A^2)_{ij} = \sum_h A_{ih} A_{hj},$$

which counts intermediate vertices. Inductively, a walk of length $k+1$ consists of a length-k walk followed by one edge; matrix multiplication counts exactly these decompositions.

If the graph is d-regular with $d > 0$, then $P = A/d$ is the transition matrix for simple random walk. Because P is symmetric and each row sums to one, the uniform distribution is stationary. If the graph is also connected and non-bipartite, its remaining eigenvalues have modulus strictly less than one:

$$1 = \lambda_1(P), \qquad \rho = \max_{2 \le j \le n} |\lambda_j(P)| < 1.$$

Expanding a probability vector in an orthonormal eigenbasis shows that its deviation from the uniform distribution decreases in Euclidean norm by at least the factor ρ^k after k steps. In a bipartite connected graph, the eigenvalue -1 causes period-two oscillation, so convergence without averaging fails.

This distinction matters: the graph's second-largest signed eigenvalue is not enough; mixing is controlled by the largest *absolute* nontrivial eigenvalue.

§1.12 Which encoding should a problem use?

Subsets and chains naturally invite binomial coefficients; independent decomposition produces products of generating functions; recursively split polygon triangulations produce Catalan numbers; convex hulls encode finite weighted averages; graph walks are encoded by powers of an adjacency matrix. The point of identifying the encoding is that it yields a calculable identity or a proof, with its conditions made explicit, rather than an analogy based on shared vocabulary.